Find the law behind the question

Adilet Search · Kazakhstan legislation

Russian and Kazakh questions → articles with sources

Daniil Talyzin · Yernur Kazhyakpar · Kuanysh Murat

Official text, a bounded snapshot

8 Tier-1 code documents · RU + KK

9,259 article-language records · 16,264 chunks

Snapshot: 17 September 2026

Test models, fusion and LLM rewriting

Does a newer embedder improve retrieval?

Does hybrid reranking help RU and KK?

Does an LLM rewrite justify its extra cost?

Newer models must earn their place

Method	Val	Test	RU test	KK test
E5 baseline	0.706	0.703	0.802	0.605
Project hybrid + reranker	0.856	0.846	0.855	0.836
Hybrid without reranking	0.778	0.770	0.816	0.724
Qwen3 Embedding 0.6B	0.539	0.548	0.667	0.430
BGE-M3	0.830	0.822	0.833	0.812
EmbeddingGemma 2 (Oct 2026)	0.710	0.716	0.802	0.630

PROXY STUDY · 1318 test queries · title-derived labels · nDCG@10

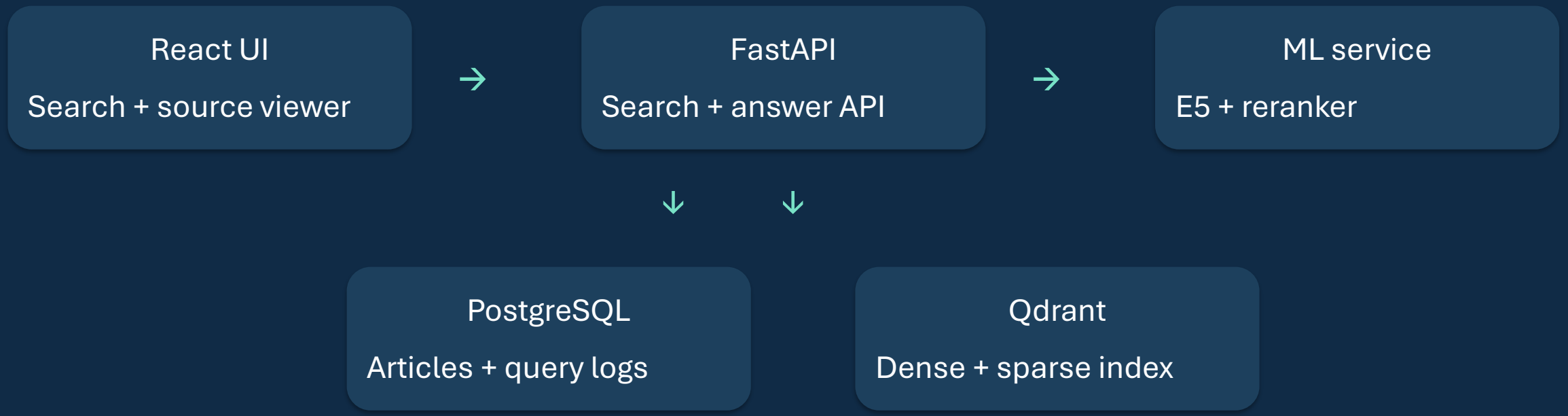
Hybrid remains the measured candidate

Hybrid – E5 test nDCG@10: +0.142

Paired 95% interval: [+0.123, +0.162]

LLM rewrite: small pilot; no benefit established

One API joins retrieval and the UI



OBSERVABILITY Prometheus → Grafana

The prototype runs in Docker

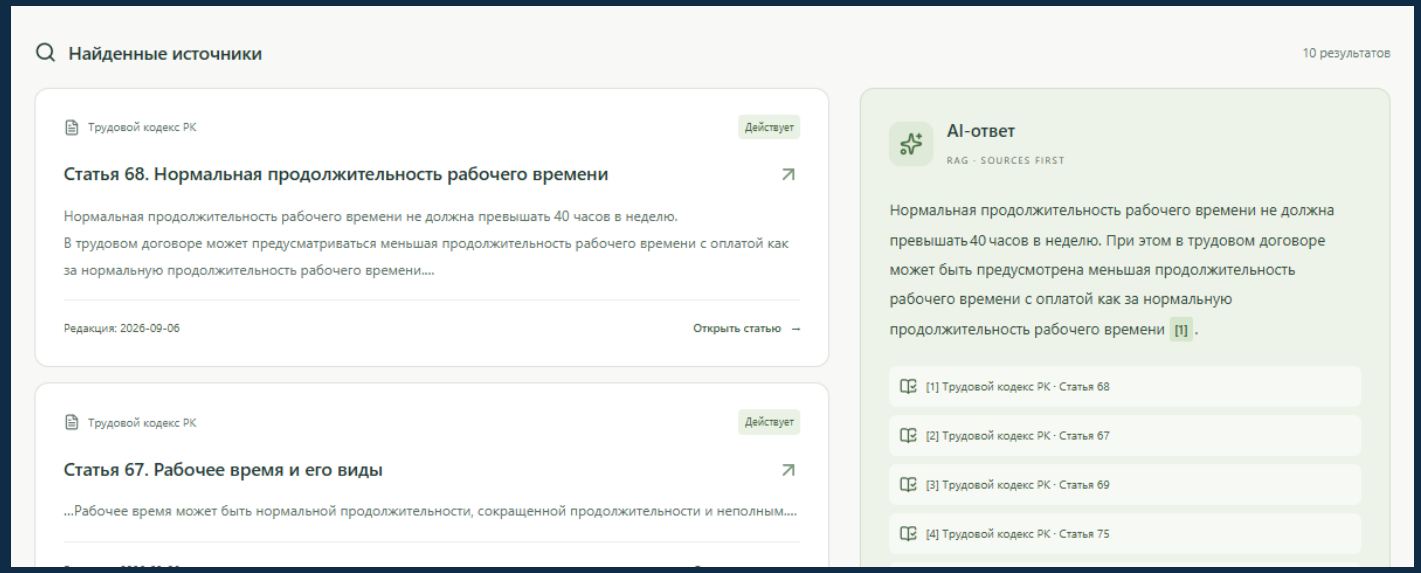
Live UI + API + real model inference

Prometheus targets up · Grafana healthy

Local access verified · public HTTPS pending

Follow the answer back to the article

- Ask about working hours or annual leave
- Inspect ranked articles and exact source text
- Switch RU ↔ KK; compare keyword and hybrid



Load testing exposes the CPU limit

100 requests · 10 concurrent users

HTTP errors: 0 · degraded: 60

Complete-response p95: 2,531 ms · target not met

The next step is stronger evidence

Independent human-labelled RU/KK questions

Long-context and prompt-sensitivity experiments

Scale LLM pilot; measure faithfulness and API load